

Jacobian diagnostics for implicit ABBA

Implicit physical step and explicit application of its Jacobian

Adaptation to the GC2D problem with Hairer's symmetric projection

1 Purpose of the diagnostics

An implicit ABBA integration can be accompanied by a diagnostic that follows two objects:

$$z_n \in \mathbb{R}^2, \quad \xi_n \in \mathbb{R}^2,$$

where z_n is the physical state and ξ_n is an infinitesimal perturbation or tangent vector. The state is computed using the complete implicit ABBA method,

$$z_{n+1} = \Psi_{h,t_n}(z_n),$$

whereas an optional observer updates the tangent vector by applying the ideal-root Jacobian of that same step:

$$\boxed{\xi_{n+1} = D\Psi_{h,t_n}(z_n) \xi_n.}$$

The responsibilities are deliberately separated:

- `ImplicitABBA1` or `ImplicitABBA2` computes the trajectory and solves for the implicit multiplier μ_n ;
- once μ_n has converged, a diagnostic observer can construct the tangent by means of a 2×2 linear system or an equivalent stage-increment recurrence.

Essential distinction

The Jacobian formula does not allow the physical step to be replaced by

$$z_{n+1} = D\Psi_{h,t_n}(z_n) z_n.$$

This equality is false for a general nonlinear map. The correct application is

$$\delta z_{n+1} = D\Psi_{h,t_n}(z_n) \delta z_n.$$

Therefore, z_{n+1} is first computed with the implicit method, and its Jacobian is then applied to ξ_n .

2 GC2D system and the small matrix w

The physical state is $z = (X, Y)^T$ and satisfies

$$\dot{z} = f(z, t) = J_0 \nabla_z \phi(z, t), \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Let $s = h/2$. We introduce the *small* 2×2 matrix

$$\boxed{w(z, t) := s D_z f(z, t) = s J_0 \nabla_z^2 \phi(z, t).}$$

For the GC2D vector field this matrix is explicitly

$$w(z, t) = s \begin{pmatrix} -\phi_{XY}(z, t) & -\phi_{YY}(z, t) \\ \phi_{XX}(z, t) & \phi_{XY}(z, t) \end{pmatrix}. \quad (1)$$

Since the Hessian of ϕ is symmetric,

$$w(z, t)^T J_0 + J_0 w(z, t) = 0.$$

The factor s is included in w so that the Jacobian of each elementary ABBA stage has the particularly simple form “identity plus one w block.”

3 Implicit ABBA physical step

For a known state z_n and a provisional value $\mu \in \mathbb{R}^2$, the two copies are initialized as

$$u^{(0)} = z_n + \mu, \quad v^{(0)} = z_n - \mu.$$

The four A – B – B – A stages are then applied:

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n), \quad (2a)$$

$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \quad (2b)$$

$$v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}), \quad (2c)$$

$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}). \quad (2d)$$

The multiplier for the step is the root $\mu_n = \mu_h(z_n)$ of the residual

$$\mathcal{R}_n(z_n, \mu) := u^{(2)}(z_n, \mu) - v^{(2)}(z_n, \mu) + 2\mu = 0. \quad (3)$$

At the exact root, the next physical state is

$$z_{n+1} = u^{(2)} + \mu_n = v^{(2)} - \mu_n. \quad (4)$$

The two expressions in (4) coincide because $\mathcal{R}_n(z_n, \mu_n) = 0$. With a finite nonlinear-solver tolerance, formulation 1 emits the neutral mean

$$\frac{1}{2} \left[(u^{(2)} + \mu_n) + (v^{(2)} - \mu_n) \right],$$

while formulation 2 emits the mean of its two simultaneous output unknowns. These representatives agree with (4) up to the nonlinear residual. Centered finite differences differentiate the emitted finite-tolerance map; the two analytic factorizations below evaluate the ideal-root tangent at the converged stages and can therefore differ from finite differences by the order of that residual.

4 Explicit calculation of the four w_i matrices

All quantities in this section are evaluated after the converged root μ_n has been obtained. The four small matrices are the matrix (1) evaluated at the four ABBA stage points:

$$w_1 := w(v^{(0)}, t_n) = s \begin{pmatrix} -\phi_{XY}(v^{(0)}, t_n) & -\phi_{YY}(v^{(0)}, t_n) \\ \phi_{XX}(v^{(0)}, t_n) & \phi_{XY}(v^{(0)}, t_n) \end{pmatrix}, \quad (5a)$$

$$w_2 := w(u^{(1)}, t_n) = s \begin{pmatrix} -\phi_{XY}(u^{(1)}, t_n) & -\phi_{YY}(u^{(1)}, t_n) \\ \phi_{XX}(u^{(1)}, t_n) & \phi_{XY}(u^{(1)}, t_n) \end{pmatrix}, \quad (5b)$$

$$w_3 := w(u^{(1)}, t_{n+1}) = s \begin{pmatrix} -\phi_{XY}(u^{(1)}, t_{n+1}) & -\phi_{YY}(u^{(1)}, t_{n+1}) \\ \phi_{XX}(u^{(1)}, t_{n+1}) & \phi_{XY}(u^{(1)}, t_{n+1}) \end{pmatrix}, \quad (5c)$$

$$w_4 := w(v^{(2)}, t_{n+1}) = s \begin{pmatrix} -\phi_{XY}(v^{(2)}, t_{n+1}) & -\phi_{YY}(v^{(2)}, t_{n+1}) \\ \phi_{XX}(v^{(2)}, t_{n+1}) & \phi_{XY}(v^{(2)}, t_{n+1}) \end{pmatrix}. \quad (5d)$$

For later use, define

$$\Sigma := w_2 + w_3. \quad (6)$$

5 Explicit calculation of the four stage Jacobians W_i

We now use a capital W_i for the 4×4 Jacobian of each elementary map in the duplicated variables (u, v) . Each one is an identity matrix plus exactly one small w_i block.

For the first A stage,

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n), \quad v^{(0)} = v^{(0)},$$

so

$$W_1 = \begin{pmatrix} I_2 & w_1 \\ 0 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & w_1 \\ 0 & 0 \end{pmatrix}. \quad (7a)$$

If $w_1 = (w_{1,ij})$, this means explicitly

$$W_1 = \begin{pmatrix} 1 & 0 & w_{1,11} & w_{1,12} \\ 0 & 1 & w_{1,21} & w_{1,22} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

For the first B stage,

$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \quad u^{(1)} = u^{(1)},$$

therefore

$$W_2 = \begin{pmatrix} I_2 & 0 \\ w_2 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & 0 \\ w_2 & 0 \end{pmatrix}. \quad (7b)$$

Equivalently,

$$W_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ w_{2,11} & w_{2,12} & 1 & 0 \\ w_{2,21} & w_{2,22} & 0 & 1 \end{pmatrix}.$$

For the second B stage,

$$v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}), \quad u^{(1)} = u^{(1)},$$

we obtain

$$W_3 = \begin{pmatrix} I_2 & 0 \\ w_3 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & 0 \\ w_3 & 0 \end{pmatrix}. \quad (7c)$$

Equivalently,

$$W_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ w_{3,11} & w_{3,12} & 1 & 0 \\ w_{3,21} & w_{3,22} & 0 & 1 \end{pmatrix}.$$

Finally, for the last A stage,

$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}), \quad v^{(2)} = v^{(2)},$$

which gives

$$W_4 = \begin{pmatrix} I_2 & w_4 \\ 0 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & w_4 \\ 0 & 0 \end{pmatrix}. \quad (7d)$$

Equivalently,

$$W_4 = \begin{pmatrix} 1 & 0 & w_{4,11} & w_{4,12} \\ 0 & 1 & w_{4,21} & w_{4,22} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Thus the Jacobian of the complete four-stage ABBA map is the ordered product

$$D\Phi_{h,t_n}^{\text{ABBA}} = W_4 W_3 W_2 W_1. \quad (8)$$

The order cannot be changed because the matrices w_i , and therefore the matrices W_i , do not commute in general.

6 Construction of the physical Jacobian

Multiplying the four matrices in (8), the Jacobian on the duplicated phase space can be written in 2×2 block form as

$$D\Phi_{h,t_n}^{\text{ABBA}} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

with

$$A = I_2 + w_4 \Sigma, \quad (9a)$$

$$B = w_1 + w_4 + w_4 \Sigma w_1, \quad (9b)$$

$$C = \Sigma, \quad (9c)$$

$$D = I_2 + \Sigma w_1. \quad (9d)$$

Notice that the factors s no longer appear separately in (9), because they are already contained inside every w_i .

To account for the implicit dependence $\mu_n = \mu_h(z_n)$, introduce the four 2×2 matrices

$$\begin{aligned} \mathcal{K} &= 4I_2 - (w_1 + w_2 + w_3 + w_4) \\ &\quad + (w_4 \Sigma + \Sigma w_1) - w_4 \Sigma w_1, \end{aligned} \quad (10)$$

$$\begin{aligned} \mathcal{L} &= w_1 - w_2 - w_3 + w_4 \\ &\quad + (w_4 \Sigma - \Sigma w_1) + w_4 \Sigma w_1, \end{aligned} \quad (11)$$

$$\mathcal{P} = I_2 + w_1 + w_4 + w_4 \Sigma + w_4 \Sigma w_1, \quad (12)$$

$$\mathcal{Q} = 2I_2 - w_1 - w_4 + w_4 \Sigma - w_4 \Sigma w_1. \quad (13)$$

Here,

$$\mathcal{K} = D_\mu \mathcal{R}_n, \quad \mathcal{L} = D_z \mathcal{R}_n.$$

Differentiating the identity

$$\mathcal{R}_n(z, \mu_h(z)) = 0$$

with respect to z gives

$$\mathcal{L} + \mathcal{K}D\mu_h(z) = 0,$$

and, provided that $\mathcal{K}$ is invertible,

$$D\mu_h(z_n) = -\mathcal{K}^{-1}\mathcal{L}.$$

Consequently, the exact Jacobian of the implicit physical step is

$$\boxed{J_n := D\Psi_{h,t_n}(z_n) = \mathcal{P} - \mathcal{Q}\mathcal{K}^{-1}\mathcal{L}.} \quad (14)$$

The matrix $\mathcal{K}$ in (10) is also the Jacobian of the residual used in Newton's method. However, $\mathcal{K}$ is not the physical Jacobian: the latter is J_n and also contains $\mathcal{P}$, $\mathcal{Q}$, and $\mathcal{L}$.

7 Applying the Jacobian without computing an inverse

In an implementation, $\mathcal{K}^{-1}$ should not be formed explicitly. To construct the entire Jacobian, solve the system with two right-hand sides:

$$\boxed{\mathcal{K}\mathcal{T} = \mathcal{L}, \quad J_n = \mathcal{P} - \mathcal{Q}\mathcal{T}.} \quad (15)$$

Since $\mathcal{K} \in \mathbb{R}^{2 \times 2}$, its factorization is inexpensive and can be reused for the two columns of $\mathcal{L}$.

7.1 Equivalent stage-increment factorization

The physical mean of the two converged copies can also be written as

$$z_{n+1} = z_n + \frac{h}{4}(k_1 + k_2 + k_3 + k_4),$$

where each k_i is the vector field evaluated at its traversed ABBA stage. Let

$$F_i = D_z f(\text{stage point}_i, \text{stage time}_i) = \frac{w_i}{s}, \quad M_\mu = D\mu_h(z_n) = -\mathcal{K}^{-1}\mathcal{L}.$$

The total stage derivatives, including the dependence of the converged multiplier on z_n , obey

$$\begin{aligned} Dk_1 &= F_1(I_2 - M_\mu), \\ Du^{(1)} &= I_2 + M_\mu + sDk_1, \\ Dk_2 &= F_2Du^{(1)}, \\ Dk_3 &= F_3Du^{(1)}, \\ Dv^{(2)} &= I_2 - M_\mu + s(Dk_2 + Dk_3), \\ Dk_4 &= F_4Dv^{(2)}. \end{aligned}$$

Consequently, a second diagnostic construction of the local physical Jacobian is

$$\boxed{J_n = I_2 + \frac{h}{4}(Dk_1 + Dk_2 + Dk_3 + Dk_4).}$$

This is an alternative computational factorization of (14), not a different numerical trajectory or a different mathematical tangent. Setting $M_\mu = 0$ would instead differentiate the explicitly averaged ABBA map.

If only a tangent vector ξ_n is to be propagated, it is not even necessary to form $\mathcal{T}$ or J_n . From (14),

$$J_n\xi_n = \mathcal{P}\xi_n - \mathcal{Q}\mathcal{K}^{-1}(\mathcal{L}\xi_n).$$

Therefore, compute successively

$$b_n = \mathcal{L}\xi_n, \quad \mathcal{K}y_n = b_n,$$

and finally

$$\boxed{\xi_{n+1} = \mathcal{P}\xi_n - \mathcal{Q}y_n.} \quad (16)$$

The only implicit part of (16) is a 2×2 linear system; no new nonlinear equation appears.

7.2 Explicit formula for the 2×2 system

Let

$$\mathcal{K} = \begin{pmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{pmatrix}, \quad b_n = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad \Delta_{\mathcal{K}} = k_{11}k_{22} - k_{12}k_{21}.$$

If $\Delta_{\mathcal{K}} \neq 0$, the solution of $\mathcal{K}y_n = b_n$ is

$$\boxed{y_n = \frac{1}{\Delta_{\mathcal{K}}} \begin{pmatrix} k_{22}b_1 - k_{12}b_2 \\ -k_{21}b_1 + k_{11}b_2 \end{pmatrix}.} \quad (17)$$

This expression makes it possible to apply (16) without a general-purpose matrix solver.

8 One-step diagnostic workflow

Given an observed implicit ABBA step with t_n , z_n , ξ_n , h , and its converged stage snapshots, the diagnostic workflow is as follows:

1. Receive the multiplier μ_n and the required stage snapshots $u^{(0)}, v^{(0)}, u^{(1)}, v^{(2)}, u^{(2)}$.
2. Evaluate explicitly w_1, w_2, w_3, w_4 using (5), and form $\Sigma = w_2 + w_3$.
3. If the full duplicated Jacobian is needed, construct the four stage matrices W_1, W_2, W_3, W_4 from (7) and multiply them in the order $D\Phi_{h,t_n}^{\text{ABBA}} = W_4W_3W_2W_1$.
4. Construct $\mathcal{K}$, $\mathcal{L}$, $\mathcal{P}$, and $\mathcal{Q}$ using (10)–(13).
5. Select either the implicit-function factorization (15), the equivalent stage-increment recurrence, or centered finite differences of the emitted map.
6. Use the resulting J_n for symplecticity analysis and optional tangent propagation $\xi_{n+1} = J_n\xi_n$.

The observer does not replace or modify the already computed physical state z_{n+1} .

9 Propagation of the accumulated Jacobian

To obtain the derivative of the numerical solution with respect to the complete initial condition, propagate a matrix $M_n \in \mathbb{R}^{2 \times 2}$ instead of a single vector. Initialize

$$M_0 = I_2$$

and, after each physical step, update it according to

$$\boxed{M_{n+1} = J_n M_n.} \quad (18)$$

Thus,

$$M_N = J_{N-1}J_{N-2} \cdots J_1J_0 = \frac{\partial z_N}{\partial z_0}.$$

The order of the products is important: the Jacobian of the most recent step always multiplies from the left.

The matrix update can also be performed without forming J_n . Solve

$$\mathcal{K}Y_n = \mathcal{L}M_n$$

for the two right-hand sides and set

$$\boxed{M_{n+1} = \mathcal{P}M_n - \mathcal{Q}Y_n.} \quad (19)$$

10 Symplecticity of the tangent propagation

If the projected ABBA step is solved exactly, its physical Jacobian satisfies

$$J_n^T J_0 J_n = J_0.$$

In two dimensions, this identity is equivalent to

$$\det J_n = 1.$$

Therefore, the application $\xi_{n+1} = J_n \xi_n$ is a symplectic linear transformation at each step. In addition, the accumulated product (18) is also symplectic:

$$M_N^T J_0 M_N = J_0, \quad \det M_N = 1.$$

In numerical computations, it is useful to monitor

$$\varepsilon_{\text{sp},n} := \left\| J_n^T J_0 J_n - J_0 \right\|, \quad \varepsilon_{\text{det},n} := |\det J_n - 1|.$$

These quantities will not be exactly zero because of roundoff, the tolerance used to solve (3), and errors in the derivatives of ϕ .

11 Interpretation in terms of two nearby trajectories

Let $z_n^\varepsilon = z_n + \varepsilon \xi_n$, with ε small. Differentiability of the implicit map gives

$$\Psi_{h,t_n}(z_n^\varepsilon) = \Psi_{h,t_n}(z_n) + \varepsilon D\Psi_{h,t_n}(z_n)\xi_n + \mathcal{O}(\varepsilon^2).$$

Since $z_{n+1} = \Psi_{h,t_n}(z_n)$ and $\xi_{n+1} = D\Psi_{h,t_n}(z_n)\xi_n$, it follows that

$$z_{n+1}^\varepsilon = z_{n+1} + \varepsilon \xi_{n+1} + \mathcal{O}(\varepsilon^2).$$

Thus, the implicit integrator computes the reference trajectory while the optional diagnostic computes the linear evolution of an infinitesimal separation from it.

12 Implementation-ready summary

At each time step, perform the two updates

$z_{n+1} = \Psi_{h,t_n}(z_n)$	using ABBA and the implicit root μ_n ,
$\xi_{n+1} = (\mathcal{P} - \mathcal{Q}\mathcal{K}^{-1}\mathcal{L})\xi_n$	using a 2×2 linear system.

The small matrices are

$$w_i = s D_z f(\text{stage point}_i, \text{stage time}_i),$$

and the four elementary 4×4 Jacobians are explicitly

$$W_1 = \begin{pmatrix} I_2 & w_1 \\ 0 & I_2 \end{pmatrix}, \quad W_2 = \begin{pmatrix} I_2 & 0 \\ w_2 & I_2 \end{pmatrix}, \quad W_3 = \begin{pmatrix} I_2 & 0 \\ w_3 & I_2 \end{pmatrix}, \quad W_4 = \begin{pmatrix} I_2 & w_4 \\ 0 & I_2 \end{pmatrix}.$$

Their ordered product is

$$D\Phi_{h,t_n}^{\text{ABBA}} = W_4 W_3 W_2 W_1.$$

The recommended tangent update remains

$$b_n = \mathcal{L}\xi_n, \quad \mathcal{K}y_n = b_n, \quad \xi_{n+1} = \mathcal{P}\xi_n - \mathcal{Q}y_n.$$

The physical state is not obtained by multiplying z_n by the Jacobian. The Jacobian is applied to the tangent vector or, equivalently, to the accumulated Jacobian with respect to the initial condition.